UCSC Extension in Silicon Valley

LINX.X409

Embedded Linux Design and Programming

3.0 Units

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Course Description

This course covers the fundamentals of building and installing a custom embedded Linux for an ARM processor platform, and provides hands-on experience for creating cross-platform environments using the GNU tools. Basic concepts for designing, testing, and customizing embedded Linux will be covered, including how the Linux scheduler is implemented, and how to write Linux kernel modules and remotely debug embedded Linux applications.

Topics include:

- An overview of embedded and real-time systems.
- Creating a cross-compiler.
- Linux device tree usage.
- Building and configuring a custom Linux kernel.
- Building and debugging Linux application source code using a GDB debugger.
- Writing kernel modules and user applications for embedded Linux using C language.
- Linux sysfs interface for GPIO.
- The basics of POSIX threads and the RTAI (real-time application interface) environment.

Prerequisite Skills

Working knowledge of C programming language and UNIX/Linux operating-system internals. Advanced C programming recommended.

Note(s): Students are expected to have access to Debian/ubuntu Linux on their computers. Options include VMware, Virtual Box, LiveCD, disk partition or separate drive. This course requires students to purchase a board (approx. $50, not included in the tuition) to complete the assignments. Students are expected to use their own Linux-based computers to do the programming project. The detailed board information will be provided in the first meeting of class.

Learning Outcomes

At the conclusion of the course, participants should be able to:

- Explain the basics of designing embedded Linux
- Master the requirements to setup a Linux cross development environment
- Use GNU tool chain to compile Linux Kernel and applications code
- Develop and download applications to run on an embedded Linux target system
- Describe the steps to write, compile and load/unload Linux Kernel modules
- Summarize the Linux File System and initramfs (Initial RAM File System)

Course Outline

Here's an outline of what is planned to be covered in class.
It may be changed based on the class's pace of instruction.

Module	Topics	Description
1	• **Embedded Linux Development Requirements.**	Course Introduction: Key Concepts & Terminology What is an Embedded System? Attributes & Examples.
2	• **Toolchains & Bootloaders**	Overview, Boot Loaders, Kernel Build, Root fs. Quiz + Assignment 1a
3	• **Yocto, Buildroot**	Concepts, Architecture, Development with Yocto, Distro Build Quiz + Assignment 1b
4	• **System Architecture, Software Updates & Prototyping**	Concepts, Storage & Flash, Memory, Core, Filesystem, Software Updates Quiz + Assignment 1c
5	• **Linux Kernel Overview**	Concepts, Module Programming, init program vs systemd, Quiz + Assignment 2a
6	• **Linux Kernel Architecture**	Proc entry, Device Trees, sysfs, interrupt handling, vs. Polling Quiz + Assignment 2b
7	• **Linux Systems Programming**	Process and Thread - Architecture & Programming, IPC Quiz + Assignment 2c
8	• **Embedded Applications**	Concepts & Overview, Embedded Python(?) Quiz + Assignment 3a
9	• **Debug Methods, Profiling - Tools & Utilities**	Concepts, Overview of Embedded Dev Tools & Debugging Quiz + Assignment 3b
10	• **Realtime Systems & Programming, Introduction**	Concepts + Understanding RT + Kernel Preemption + Measuring Scheduling Latencies (Transition topic to RTES, Intro Track: **EMBD.X410**) Quiz + Assignment 3c

Required Tools and Materials

- Raspberry Pi 3B+

Reference Materials

- *Embedded Linux Programming:* **Frank Vasquez, Chris Simmonds** (Authors).
  Packt (Publisher); ISBN-13: 978-1-789953-038-4; ISBN-10: 9781789530384
- *Exploring Raspberry Pi: Interfacing to the Real World with Embedded Linux*, **Derek Molloy** (Author).
  Wiley (Publisher); ISBN-13: 978-1119188681 ISBN-10: 1119188687

Additional Reference Materials

- *Embedded Linux Primer,* **Christopher Hallinan**(Author). (Dated Reference).
  Prentice Hall (Publisher) ISBN-13: 978-0-137-01783-6
- *An Embedded Software Primer,* **David E. Simon** (Author). (Dated Reference).
  Addison-Wesley Professional (Publisher); ISBN-13: 978-0201615692; ISBN-10: 020161569X.
- *Introduction to Embedded Systems: A Cyber-Physical Systems Approach,*
  Edward Ashford Lee, **Sanjit Arunkumar Seshia** (Authors).
  The MIT Press (Publisher); ISBN-13: 978-0262533812; ISBN-10: 0262533812
- *Principles of Cyber-Physical Systems,* **Rajeev Alur** (Author).
  The MIT Press (Publisher); ISBN-13: 978-0262029117; ISBN-10: 0262029111
- *Programming Embedded Systems*, **Michael Barr** (Author). (Dated Reference).
  O'Reilly Media (Publisher); ISBN-13: 978-1565923546; ISBN-10: 1565923545

Performance Evaluation

Activity	%age	Description
10 Quizzes	60%	Applying lecture knowledge and skills
3 Assignments	40%	Projects
Total:	**100%**	
Extra Credit	**10%**	Attendance, Class Participation, Coursework Submission Regularity

Grading

Letter grades (A through F) are the default options. However, students have until the day before the course end date to change their grading preference to a Credit/No Credit Option.

Grading scale

Click Here to Review the Grading and Credits URL.

Grade options	%
A, A-	≥ 90
B, B- to B+	80-90
C, C- to C+	70-80
D, D- to D+	60-70
F	59 and below
Credit	60 and above
No Credit	59 and below

***For alternative grading options, students MUST contact extensiongrades@ucsc.edu using the Alternative Grade Form.**